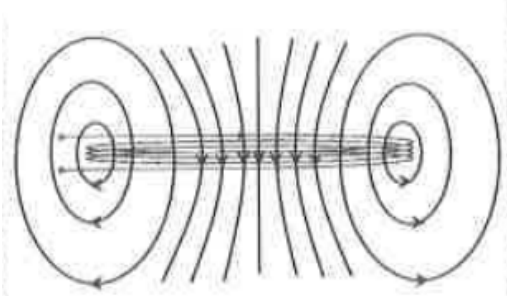


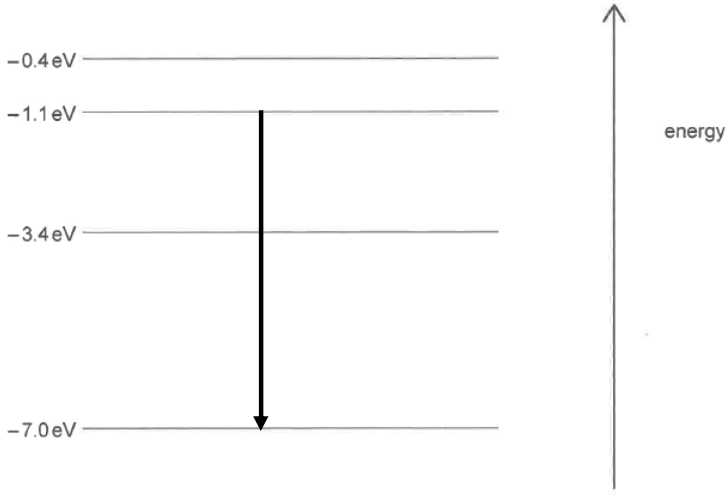
$$= 599.45 = 600 \text{ V}$$

|                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>4(c)(ii)</b> | Fig.2 shows the scalar sum of electric potential due to both A and B. Since A and B are of the same signs, the electric potential in Fig.2 must be higher than (c)(i).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>5(a)</b>     | <p>When the switch is opened, the reading is zero.</p> <p>When the switch is closed, the reading is non-zero for a short time.</p> <p>When the switch is reopened, the reading change sign and is non-zero for a short time. But the reading lasts for a short time and becomes zero again.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>5(b)</b>     | <p>When the switch is closed, current flows through the primary coil and create a magnetic field. Magnetic flux lines are linked from the primary coil to the secondary coil by the core. There is an increase in magnetic flux linkage with time in the secondary coil. By Faraday's law, e.m.f is induced in the secondary coil.</p> <p>When the switch is reopened, no current flows through the primary coil and hence no magnetic field. There is a decrease in magnetic flux linkage with time in the secondary coil. By Faraday's law, e.m.f of opposite polarity is induced in the secondary coil.</p> <p>But the reading becomes zero again after the switch is opened. No current flows and hence there is no magnetic field and there is no change of magnetic flux linkage with time in the secondary coil. Hence, no e.m.f induced.</p> |

| No.              | Solution                                                                                                                                                                                                                                                                                                  |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>6(a)</b>      | From the graph, $R$ at $20^{\circ}\text{C} = 60\ \Omega$<br>By potential divider principle,<br>$0.43 = \frac{r}{r + 60} \times 1.50$ $0.43(r + 60) = 1.50r$ $1.07r = 25.8$ $r = \frac{25.8}{1.07} = 24.112 = 24\ \Omega \text{ (shown)}$                                                                  |
| <b>6(b)</b>      | From the graph, $R$ at $32^{\circ}\text{C} = 40\ \Omega$<br>$R_{\text{eff}} = \left( \frac{1}{24} + \frac{1}{40} \right)^{-1} = 15\ \Omega$ $I = \frac{\mathcal{E}}{R_{\text{eff}}} = \frac{1.50}{15} = 0.10\ \text{A}$                                                                                   |
| <b>7(a)</b>      | They are waves of the same frequency leading to a constant phase difference between them.                                                                                                                                                                                                                 |
| <b>7(b)</b>      | $d \sin \theta = n\lambda$ $\left( \frac{0.001}{300} \right) \sin \theta = 2(640 \times 10^{-9}) \Rightarrow \theta \approx 22.6^{\circ}$ $\tan \theta = \frac{y}{2.1} \Rightarrow y = 2.1 \times \tan(22.6^{\circ}) \approx 0.873\ \text{m}$                                                             |
| <b>8(a)(i)</b>   | Electric field strength at a point is the electric force per unit positive charge placed at that point.                                                                                                                                                                                                   |
| <b>8(a)(ii)</b>  | To prevent electrons from colliding with other molecules, as scattered electrons will lose too much kinetic energy, resulting in fewer electrons exiting from the anode with an extremely wide range of kinetic energy.                                                                                   |
| <b>8(a)(iii)</b> | Gain in Kinetic energy = Loss in electric potential energy<br>$2.48 \times 10^{-16} = e\Delta V$ $\Delta V = \frac{2.48 \times 10^{-16}}{1.60 \times 10^{-19}} = 1550\ \text{V}$                                                                                                                          |
| <b>8(a)(iv)</b>  | Electrons are emitted from the cathode in different directions. The direction of electric field is horizontal and will only affect the horizontal component of the velocity of the electrons while the vertical component of velocity remained unchanged, hence the resultant velocity will be different. |
| <b>8(b)(i)</b>   | $B = \frac{\mu_0 NI}{2r} = \frac{4\pi \times 10^{-7} \times 120 \times 3.5}{2(0.15)} = 0.00176\ \text{T}$                                                                                                                                                                                                 |
| <b>8(b)(ii)</b>  |                                                                                                                                                                                                                        |

| No.             | Solution                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>8(c)(i)</b>  | The velocity of electron is always perpendicular to the magnetic field. By Fleming's left hand rule, a magnetic force always acts perpendicular to both velocity and magnetic field. (i.e. towards a common centre) and the speed of the electron remains unchanged. Hence this magnetic force provides the centripetal force for the electron to move in a circular path.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>8(c)(ii)</b> | <p>Magnetic force provide the centripetal force for the electron to move in circular motion.</p> $Bqv = \frac{mv^2}{r}$ $r = \frac{mv}{Bq}$ $= \frac{9.11 \times 10^{-31} \times 2.40 \times 10^7}{1.43 \times 0.0017593 \times 1.60 \times 10^{-19}}$ $= 0.054317 = 0.0543 \text{ m}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>8(d)</b>     | <p>The velocity of the electron can be resolved into 2 directions: parallel (e.g. <math>v \cos \theta</math>) and perpendicular (e.g. <math>v \sin \theta</math>) to the magnetic field.</p> <p>By Fleming's left hand rule, a magnetic force always acts perpendicular to both <math>v \sin \theta</math> and magnetic field. This provide centripetal force for the electron to move in a circular path in a plane perpendicular to the magnetic field.</p> <p>As <math>v \cos \theta</math> is parallel to the magnetic field, there will be no magnetic force and the electron will move parallel to the magnetic field.</p> <p>Both motion happened at the same time and hence its moves in a helical path.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>9(a)</b>     | <p>In particulate nature, light is considered as discrete packets of energy called photon. Each photon has energy <math>E = hf</math>. Since one photon can only interact with one electron, the photons caused emission of photoelectrons whose K.E. depends on the energy provide by each photon. Since the frequency <math>f</math> in observation 1 is constant, the energy of each photon is constant and thus energy supplied to each photoelectron is constant. It does not depend on the intensity of the incident radiation as intensity only affect the number of photons incident per unit time on the metal and not the energy of each photons. The first observation does not support wave nature of electromagnetic radiation as wave theory states that at higher intensity, the higher power per unit area should cause photoelectrons of higher K.E. to be emitted, which was not observed in experiments.</p> <p>For observation 2, if frequency of the photon is below a threshold frequency, energy of each photon is not enough to remove a electron from the metal surface as its energy is lower than the minimum energy needed. This again does not support wave nature of electromagnetic radiation as with higher intensity, you expects the electrons in the metal to sooner or later accumulate sufficient energy over time to get the minimum energy needed to be emitted. Again, this was not observed in experiments.</p> |
| <b>9(b)</b>     | Threshold frequency.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

| No.                         | Solution                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>9(c)</b>                 | $KE_{max} = hf - W_{min}$ $= \frac{hc}{\lambda} - \phi$ $= \frac{(6.63 \times 10^{-34})(3.0 \times 10^8)}{210 \times 10^{-9}} - (1.60 \times 10^{-19})(4.33)$ $= 2.54 \times 10^{-19} \text{ J}$                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>9(d)</b>                 | $KE_{max} = eV_s$ $V_s = \frac{KE_{max}}{e}$ $= \frac{2.54 \times 10^{-19}}{1.60 \times 10^{-19}}$ $= 1.59 \text{ V}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>9(e)(i), (ii), (iii)</b> | <div data-bbox="451 783 1084 1129" data-label="Figure"> </div> <p><b>(e)(ii)</b> shorter wavelength means photons have higher frequency. Thus the incoming photons have higher energy resulting in emitted electrons having higher K.E. and thus higher stopping potential. Since intensity is the same, number of photons emitted per unit time by the source is lower resulting in lower number of photoelectrons emitted and thus a lower current.</p> <p><b>(e)(iii)</b> with greater work function energy, the maximum K.E. of emitted electrons is lower and thus, the stopping potential is lower.</p> |
| <b>9(e)(iv)</b>             | <p>The photoelectrons are emitted with varying K.E. At <math>V</math> values equal to or greater than the stopping potential, the electrode is negative with respect to the zinc plate and at these potentials, photoelectrons emitted with the maximum K.E. possible will have insufficient K.E. to reach the electrode as all the K.E. is already converted to electric potential energies before reaching the electrode. Since no electrons reaches the electrode and current is the rate of charge flow, the current is zero.</p>                                                                         |

| No.      | Solution                                                                                                                                                                                                                                                                                                                                                                                                                                 |
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| 9(f)(i)  |  <p>Energy of the 210 nm electromagnetic radiation from (c) is</p> $E = \frac{hc}{\lambda}$ $= \frac{(6.63 \times 10^{-34})(3.0 \times 10^8)}{210 \times 10^{-9}}$ $= 9.47 \times 10^{-19} \text{ J}$ $= 5.92 \text{ eV}$ <p>Thus, this will be due to emission between energy level <math>-1.1 \text{ eV}</math> to <math>-7.0 \text{ eV}</math>.</p> |
| 9(f)(ii) | <p>Longest wavelength emitted through transition with the smallest energy difference.</p> $(-0.4 - (-1.1))(1.6 \times 10^{-19}) = \frac{hc}{\lambda}$ $\lambda = \frac{(6.63 \times 10^{-34})(3.0 \times 10^8)}{(-0.4 - (-1.1))(1.6 \times 10^{-19})}$ $= 1.78 \times 10^{-6} \text{ m}$                                                                                                                                                 |